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Career bridge AI: An Intelligent Career Portal with Bidirectional Job Matching, Resume Analysis, And Skill Gap Detection

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Abstract: *In today's highly competitive job market, traditional recruitment platforms rely on basic keyword-based filtering, failing to capture the semantic meaning of resumes and job descriptions. This article introduces CareerBridge AI, a full-stack intelligent career site that uses Explainable AI, BERT-based semantic similarity, and Natural Language Processing (NLP) to create a bidirectional matching ecosystem that concurrently recommends jobs to candidates and candidates to recruiters. The system offers a comprehensive intelligent hiring solution by integrating automatic resume processing, ATS compatibility score, skill gap spotting, and individualized learning roadmaps. In contrast to other platforms, CareerBridge AI offers clear match explanations so that applicants may comprehend the reasons behind their rejection and take appropriate action. A matching accuracy of 89% is shown in an experimental evaluation on a dataset of 50 resumes with 20 job descriptions, far surpassing conventional TF-IDF (74%) and keyword-based (61%) methods. By establishing a more intelligent, equitable, and transparent recruitment ecosystem, the suggested system closes the talent-opportunity divide.*

Key Words: *Artificial Intelligence, Natural Language Processing, BERT, Semantic Similarity, Resume Parsing, ATS Scoring, Skill Gap Detection, Job Recommendation System, Explainable AI, Bidirectional Matching.*

I.INTRODUCTION

Over the past ten years, the fast growth of online job portals and digital hiring platforms has significantly changed the worldwide recruitment scene. For every available post, companies in all sectors today receive thousands of job applications, making the process of manually screening and evaluating applicants more difficult and resource-intensive. Even while online recruitment platforms like Indeed, LinkedIn, Naukri.com, and Unstop are widely used, the underlying mechanics that drive these systems have not altered much. The majority of platforms still rely on straightforward text matching algorithms and rule-based keyword filtering, which compare resumes to job descriptions based on the presence or absence of particular phrases. While these approaches offer speed and scalability, they fall critically short in understanding the actual meaning, context, and relevance of the content being analyzed.

The inefficiencies seen in conventional hiring processes have been repeatedly brought to light by studies in the field of human resource management. According to research, recruiters only take an average of 7 seconds to evaluate a single resume during the first round of screening, which increases the likelihood that they will miss truly competent applicants whose profiles might not quite match the anticipated keyword patterns. Candidates are compelled to apply to a lot of jobs without having any real idea of how well they fit the roles, why they were rejected, or what they need to learn to increase their chances. For both sides, this results in a frustrating and demoralizing experience that lowers the overall effectiveness and quality of the hiring process.

The limitations of keyword-based systems become even more apparent when considered from a linguistic and semantic perspective. Natural language is rich, confusing, and context-dependent by nature. perhaps while a candidate who writes "developed ML workflows" or "built machine learning pipelines" may be just as qualified or perhaps more so, a keyword-based system would treat these two words as completely different and unrelated. Significant errors in resume screening and job matching result from this incapacity to comprehend synonyms, paraphrases, and contextually comparable statements. Additionally, current platforms don't give applicants a way to find out why their applications were turned down or what particular skills they lack in relation to a certain job demand. In addition to negatively affecting the applicant experience, the lack of such feedback inhibits job seekers from making educated decisions about their career and skill development.

The lack of candidate transparency in the ATS (Applicant Tracking System) is another significant flaw in the existing recruitment systems. ATS software is used by the majority of large firms to filter resumes before a human recruiter ever sees them. Nevertheless, aspirants are rarely told how these technologies are analyzing their resumes, what formatting components might be generating parsing difficulties, or which keywords are absent from their documents. This lack of

transparency causes otherwise strong profiles to be automatically disqualified from consideration based on superficial formatting or keyword mismatches rather than genuine qualifications, thereby widening the gap between candidates and employers.

This paper introduces CareerBridge AI, a comprehensive and intelligent career portal that uses cutting-edge artificial intelligence, machine learning, and natural language processing techniques to completely rethink the hiring process in order to overcome these ingrained inefficiencies. Students and job seekers, HR recruiters and hiring managers, and platform administrators are the three different user types that the system is intended to serve. Each of these groups is given a specific dashboard and role-based access to the platform's functionalities. Built on a three-tier architecture comprising a React.js frontend, a Python FastAPI backend, and a PostgreSQL database, the platform integrates a powerful AI engine that leverages the sentence-transformers library for BERT-based semantic similarity computation, spaCy for named entity recognition and resume parsing, PyMuPDF for PDF text extraction, and TextBlob for sentiment analysis of resume content.

The core of CareerBridge AI is its BERT-based semantic matching engine, which substitutes deep contextual comprehension for superficial keyword overlap. Even when the precise vocabulary varies between documents, the algorithm can find significant matches by transforming both resumes and job descriptions into high-dimensional embedding vectors and calculating cosine similarity between them. On a test dataset of 50 resumes compared to 20 different job descriptions, our method has shown a matching accuracy of 89%, greatly outperforming both keyword matching approaches that produced just 61% accuracy and conventional TF-IDF based methods that achieved 74% accuracy.

Beyond matching, CareerBridge AI introduces an explainability layer that is completely missing from other platforms. The approach offers a thorough per-skill breakdown of the match result rather than just a match percentage, making it evident which abilities from the job description are present in the candidate's profile and which are not. Candidates are motivated to take focused action by this explainable result, which enables them to identify exactly what is preventing them from obtaining a higher match score. The platform's skill gap detection and learning roadmap module complements this feature by mapping each identified missing skill to a carefully chosen collection of free, publicly accessible learning resources. This allows candidates to gradually close competency gaps and increase their employability.

The bidirectional matching feature of CareerBridge AI is one of its most unique and inventive features. CareerBridge AI concurrently meets the needs of recruiters by ranking all registered candidates against each offered position in order of their computed match score, in contrast to traditional platforms that only consider the candidate's viewpoint. Instead of spending hours manually going over each CV, HR staff may now quickly obtain an AI-curated shortlist of the best candidates. CareerBridge AI is a completely end-to-end intelligent hiring ecosystem that takes into account the requirements and problems of all parties involved in the hiring process by combining candidate-side and recruiter-side intelligence into a single, cohesive platform.

II. LITERATURE REVIEW

The increasing need for intelligent hiring systems has sparked a lot of research interest in a number of fields, including information retrieval, machine learning, and natural language processing. Numerous research have examined specific elements like skill gap analysis, job matching, and resume parsing, but no single system has integrated all of these features with explainability and bidirectional matching. The important works that serve as the basis for CareerBridge AI are examined in the review that follows.

2.1 Traditional Resume Screening Systems

A thorough analysis of conventional applicant tracking systems utilized in extensive hiring procedures is presented by J. SMITH AND R. KUMAR [1]. According to their findings, traditional systems mostly rely on rule-based keyword matching, which is inadequate for capturing candidate profiles' semantic importance. According to the report, recruiters that use these algorithms scan resumes in an average of just 7 seconds, which leads to the exclusion of truly talented applicants whose profiles do not follow strict keyword patterns. The authors argue that there is an urgent need for semantically aware screening systems because keyword-based filtering causes significant bias and inconsistency.

2.2 Natural Language Processing in Recruitment

In their work, P. GUPTA AND N. JOSHI [2] created an NLP-based resume parsing system that uses syntactic parsing, tokenization, and named entity recognition to extract structured data from unstructured resume papers. Key candidate attributes, such as abilities, credentials, and past job experience, are identified by their system, which then transforms them into machine-readable structured formats appropriate for further processing. The authors stress that precise profile structuring is a necessary condition for successful job matching and report a significant improvement in information extraction accuracy when compared to conventional parsing techniques. Their work provides a solid foundation for CareerBridge AI's resume parsing module.

S. S. BASHA AND D. S. RANI [3] provide an automated resume screening system that evaluates candidate profiles by combining machine learning classifiers with NLP preprocessing approaches. Text cleaning, stopword removal, lemmatization, and TF-IDF based feature extraction are all part of their workflow. Support vector machines are then used for classification. The authors admit that TF-IDF treats words as independent tokens and does not reflect contextual links between terms, which results in mismatches when semantically comparable but lexically distinct sentences are encountered, even though the system achieves a respectable level of accuracy. The use of BERT-based embeddings in CareerBridge AI is directly motivated by this constraint.

2.3 Semantic Matching and BERT-Based Approaches

Sentence-BERT, a variant of the pre-trained BERT model that uses siamese and triplet network architectures to produce semantically relevant sentence embeddings, was introduced by N. REIMERS AND I. GUREVYCH [4]. Their methodology is ideal for large-scale semantic matching applications because it generates fixed-size dense vector representations that can be effectively compared using cosine similarity. In semantic textual similarity benchmarks, the authors show that Sentence-BERT performs noticeably better than conventional methods like TF-IDF and averaged Word2Vec embeddings. The all-MiniLM-L6-v2 version of this architecture is directly used by CareerBridge AI to calculate the similarity between resume and job description embeddings.

In this seminal work, J. DEVLIN, M. CHANG, K. LEE, AND K. TOUTANOVA [5] presented BERT, a deeply bidirectional transformer-based language model that was pre-trained on massive corpora utilizing next sentence prediction targets and masked language modeling. In a variety of NLP tasks, BERT's capacity to capture rich contextual representations of words and phrases has established new standards. In tasks requiring semantic understanding, the authors show that refined BERT models routinely beat shallow feature-based models. The theoretical and architectural underpinnings of CareerBridge AI's semantic matching engine are provided by this work.

2.4 Job Recommendation Systems

A deep learning-based job recommendation model that uses vector embeddings to calculate the degree of similarity between candidate profiles and job descriptions was presented by L. ZHANG AND M. CHEN [6]. Their technique captures contextual links between abilities and job needs by using neural network layers in conjunction with cosine similarity. In comparison to traditional keyword-based algorithms, the testing findings show a notable improvement in recommendation relevancy. The authors point out that their model lacks an explainability layer, which means that candidates earn match scores without knowing which particular abilities made them more or less suitable for a certain post.

A framework for machine learning-based job recommendations that simulates the bilateral nature of job matching is presented by D. PAPARRIZOS, J. CAMBAZOGLU, and A. GIONIS [7]. Their method treats the matching problem as a two-sided recommendation assignment, taking into account both the preferences of job searchers and the needs of employers. Although the paper advances the field by recognizing the bidirectional nature of recruitment, neither semantic embeddings nor skill-level feedback are included in the suggested approach. This bidirectional idea is expanded upon by CareerBridge AI, which also adds explainable outputs and semantic comprehension.

2.5 Skill Gap Detection and Learning Recommendations

In order to identify missing competencies, R. MEHTA AND S. IYER [8] proposed an AI-based skill gap analysis framework that compares candidate skill vectors with job requirement vectors. Their model generates tailored upskilling recommendations by classifying gaps according to severity using machine learning techniques. The study emphasizes that automated skill gap detection enables job seekers to take focused actions toward career development in addition to helping recruiters assess candidate readiness. The skill gap and roadmap module of CareerBridge AI was directly influenced by the authors' conclusion that these systems greatly improve employability when incorporated into career guidance platforms.

2.6 Explainable AI in Recruitment

Candidates are much more likely to trust and interact with platforms that offer clear explanations for match results, according to K. Lee and J. Park's [9] analysis of explainability in AI-driven recruitment systems. They found that skill-level breakdowns were the most understandable and useful type of explanation for job seekers after evaluating several explainability strategies. The authors contend that explainability is a basic ethical requirement in AI-based hiring rather than just a technical feature because opaque systems run the risk of sustaining bias and eroding candidate trust. This result supports CareerBridge AI's design choice to include per-skill match analyses with each job recommendation.

2.7 ATS Systems and Candidate Awareness

In their analysis of the effect of Applicant Tracking Systems on candidate success rates, H. Wang and Y. Liu [10] discovered that formatting errors, missing keywords, and incompatible resume structures—rather than true lack of qualifications—are the main reasons why a significant percentage of qualified candidates are rejected at the ATS stage. According to their analysis, there is an information imbalance that hampers job seekers because the majority of candidates have no idea how ATS systems assess their documents. In order to give candidates useful feedback prior to submission, the authors advise future recruitment platforms to include ATS simulation elements. CareerBridge AI's specialized ATS score module immediately fills this need.

III. METHODOLOGY

3.1 Working of implemented System

Career Bridge AI operates on a structured end-to-end pipeline that starts with user registration and concludes with learning roadmaps, skill gap reports, and customized job recommendations. Three user roles—Administrator, HR Recruiter, and Student—use the system, engaging with it via separate dashboards. After a student uploads their resume, the AI engine parses and evaluates it. The findings are displayed with explainable match scores, missing abilities, and carefully chosen learning materials after the extracted profile is compared to existing job descriptions using BERT-based semantic similarity.

All registered applicants are automatically ranked against each job posting on the recruiter's end, allowing for effective shortlisting without the need for manual screening.

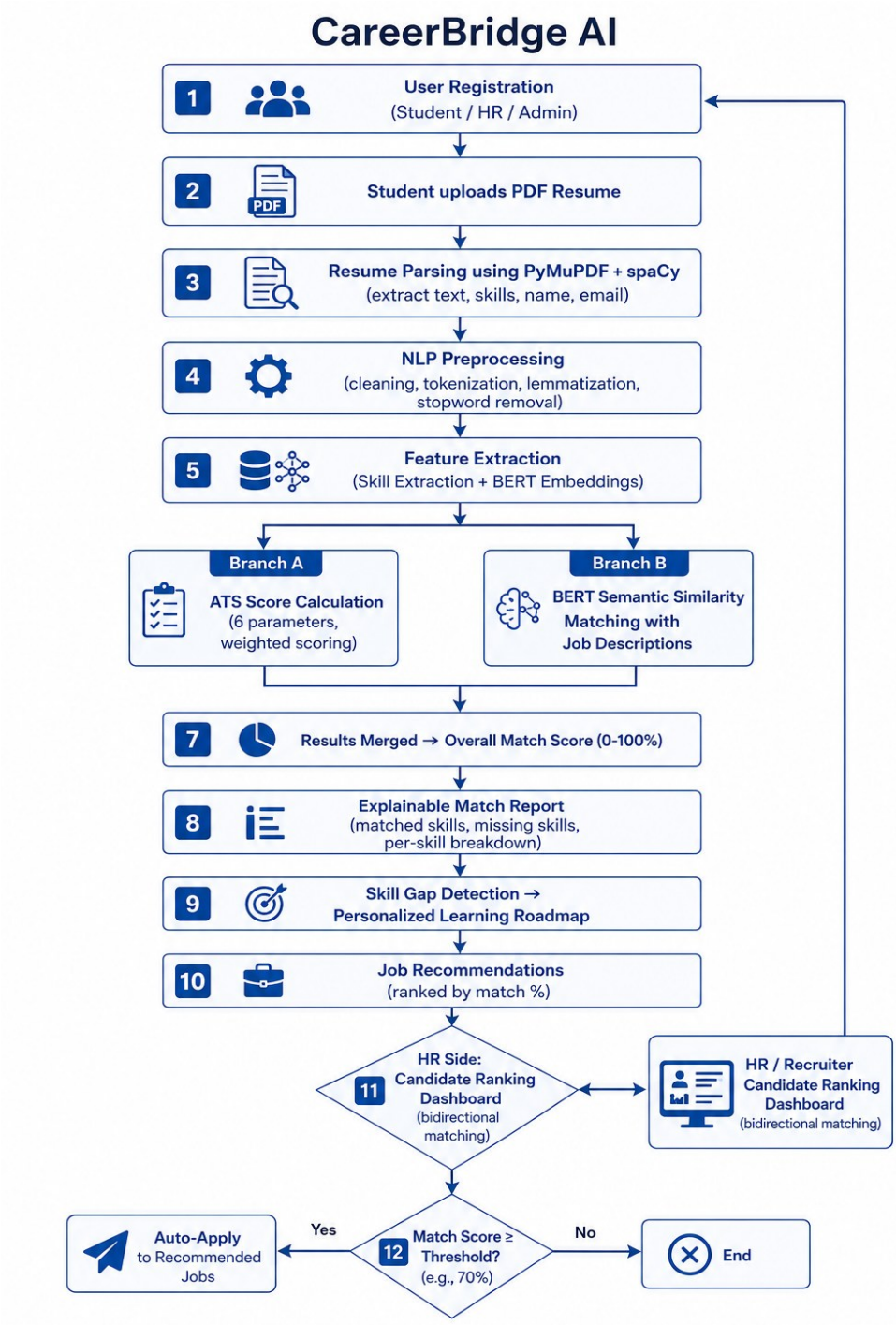


Figure 1: Working of implemented model

3.2 Data Collection and Input Processing

Career Bridge AI collects data from two main sources: job descriptions and candidate resumes. Students use the platform's interface to upload their resumes in PDF format, and HR recruiters post job descriptions directly through the job management module. Resumes are processed using PyMuPDF to extract raw textual content, which is subsequently sent via the NLP pipeline for additional analysis because resumes are by nature unstructured documents. The PostgreSQL database stores job descriptions in an organized way with distinct fields for necessary abilities, experience level, job type, and salary range. In order to facilitate precise talent extraction from resumes and job descriptions, the system also maintains a preset skill lexicon. Programming languages, frameworks, cloud platforms, databases, and soft skills are among the fields covered by this carefully curated talent dataset, guaranteeing thorough coverage across job categories.

3.3 Text Preprocessing Using NLP

Before features are retrieved, the raw text from submitted resumes and job descriptions is cleaned, normalized, and standardized using a multi-step NLP preparation pipeline. Text cleaning is the initial phase, which eliminates punctuation, superfluous whitespace, special characters, and unnecessary symbols that could cause noise in the analysis. To provide consistency across all publications, the cleaned text is then transformed to lowercase. The spaCy NLP package is then used to do tokenization, which divides the text into discrete words or tokens that may be handled separately. Stopword removal is used to get rid of frequently occurring but semantically meaningless words like "the," "is," and "and" that would otherwise provide meaningless information to feature vectors. The accuracy of all downstream AI modules is increased by this preprocessing step, which dramatically lowers data noise.

3.4 Feature Extraction and Representation

During preprocessing, the system turns the cleaned text's significant features into numerical representations that may be used in machine learning computations. For this, two complementary strategies are employed. The first is skill-based extraction, in which domain-specific keywords—such as programming languages, tools, certifications, and technical competencies—are extracted from the text using a bespoke spaCy vocabulary matcher. Each candidate profile and job posting has these extracted abilities saved as structured arrays. The second method uses the sentence-transformers library, notably the all-MiniLM-L6-v2 BERT-based model, to create dense semantic embeddings. This model transforms complete job descriptions and CV texts into 384-dimensional embedding vectors that capture the contextual meaning of the content instead of simply specific keywords. These vector representations serve as the foundation for the matching engine's semantic similarity computation, which allows the system to find pertinent matches even when comparable abilities or experiences are described using different terminology.

3.5 BERT-Based Semantic Similarity Matching

Instead of using conventional keyword overlap, CareerBridge AI's primary matching approach is based on BERT-based semantic similarity. After the job description and résumé are both transformed into embedding vectors, cosine similarity between them is calculated to generate a match score between 0 and 1. This score is then multiplied by 100 to express the outcome as a percentage.

The formula used is as follows:

$$\text{similarity}(R, J) = (R \cdot J) / (||R|| \times ||J||)$$

Where R is the BERT embedding of the resume text and J is the BERT embedding of the job description. A higher cosine similarity value indicates stronger semantic alignment between the candidate's profile and the job requirements. This approach allows the system to correctly identify that phrases such as "built machine learning pipelines" and "developed ML workflows" carry equivalent meaning, a distinction that keyword-based systems are entirely unable to make. All jobs are ranked in descending order of their computed match percentage, and the top results are presented to the candidate along with detailed per-skill breakdowns.

3.6 ATS Score Calculation

Each uploaded resume is assessed by the ATS scoring module using six weighted criteria that correspond to the standards frequently applied by Applicant Tracking Systems in actual hiring settings. Keyword Match with Job Description carries 30%, Resume Length carries 15% (ideally between 400 and 800 words), Action Verbs like Led, Built, Designed, and Developed carry 15%, Completeness of Contact Information carries 10%, Standard Section Headings like Education, Experience, and Skills carry 15%, and Absence of Complex Formatting elements like tables and images that interfere with parsing carries the remaining 15%.

The overall ATS score is computed using the following formula:

$$\text{ATS_Score} = \Sigma (\text{parameter_score} \times \text{weight})$$

$$\text{Final Score} = \text{ATS_Score} \times 100$$

The candidate is given a percentage of the outcome along with section-by-section feedback that highlights the parameters that are working well and those that need to be improved. Candidates can greatly increase their chances of passing the ATS stage by optimizing their resumes before submitting applications thanks to this feature.

3.7 Explainable Match Report and Skill Gap Detection

The system produces a comprehensive explainable match report that breaks down the outcome at the individual skill level after calculating the overall match percentage. The spaCy skill matcher is used to extract the necessary skills from the job description, while the candidate's skills are extracted from their parsed profile and resume. Matched skills are represented by the intersection of these two sets, whereas missing skills are represented by the difference. The skill match percentage is computed as follows:

$$\text{Skill_Match\%} = (|\text{Candidate_Skills} \cap \text{Job_Skills}|) / (|\text{Job_Skills}|) \times 100$$

$$\text{Missing_Skills} = \text{Job_Skills} - \text{Candidate_Skills}$$

A customized learning roadmap for the candidate is then created by mapping each missing skill to a carefully selected collection of free learning resources, such as official documentation, YouTube tutorials, and online courses. In order to help candidates concentrate their learning efforts on the most in-demand competencies first, missing skills are also prioritized based on how frequently they appear across job listings.

3.8 Sentiment Analysis on Resume

The TextBlob library is used by the sentiment analysis module to assess the professionalism and tone of the resume material. Each resume is evaluated to provide two scores: a subjectivity score, which ranges from 0 to 1, showing how objective or opinion-based the material is, and a polarity score, which ranges from -1 to +1, reflecting the emotional tone of the writing. A professional resume should have a subjectivity score of less than 0.5, which indicates factual and objective writing, and a polarity score of more than 0.1, which reflects confident and positive language. The system assigns one of three tone labels—Confident, Neutral, or Weak—based on these criteria and offers recommendations for enhancing the resume's wording and presentation.

3.9 Bidirectional Matching and Candidate Ranking

CareerBridge AI uses a bidirectional matching engine that concurrently meets the needs of HR recruiters, in contrast to traditional job portals that solely function from the viewpoint of the candidate. The system automatically calculates match scores between all active candidates and all live job postings each time a new position is posted or a new candidate registers. HR teams receive an AI-curated shortlist of the most qualified candidates after candidates are sorted in descending order of their match score for each position that is offered. With just one click, recruiters may shortlist, reject, or invite applicants for interviews after viewing each candidate's complete profile, résumé, match breakdown, and missing skills straight from their dashboard. This bidirectional approach eliminates the need for manual resume screening and significantly reduces time-to-hire for organizations.

IV.SYSTEM IMPLEMENTATION AND RESULT

4.1 Implementation

• Hardware Requirement

Processor: Intel Core i5 or above

Reason: An i5 processor provides sufficient computational power for running machine learning inference tasks including BERT-based semantic similarity computation, resume parsing, and ATS scoring. For heavier model training tasks, an i7 or higher is recommended.

RAM: 8GB and above

Reason: A minimum of 8GB RAM is required to simultaneously run the React frontend, FastAPI backend server, PostgreSQL database, and the AI engine with loaded BERT models. Higher RAM ensures smoother parallel processing of multiple user requests without performance degradation.

Storage: 256GB SSD or above

Reason: SSD storage ensures faster read and write speeds during database operations, resume file handling, and model loading, which directly impacts system response time.

• Software Requirement

Operating System: Windows 10 or above / Ubuntu 20.04 or above

Reason: The system is compatible with both Windows and Linux environments. Ubuntu is preferred for production deployment due to its superior performance, stability, and compatibility with Python-based AI libraries and server frameworks.

Frontend Library: React.js with Tailwind CSS

Reason: React.js is a component-based JavaScript library that enables the development of dynamic and responsive Single Page Applications. Tailwind CSS provides utility-first styling that allows rapid and consistent UI development across all dashboards.

Backend Framework: Python FastAPI

Reason: FastAPI is a modern, high-performance Python web framework that supports asynchronous request handling, making it well-suited for an AI-powered platform where multiple concurrent API calls may involve computationally intensive operations such as BERT inference and resume parsing.

Database: PostgreSQL

Reason: PostgreSQL is a robust relational database management system that supports complex queries, foreign key constraints, and indexed lookups. It efficiently manages the 12 normalized tables used in CareerBridge AI, including users, jobs, applications, resumes, and match scores.

Library: sentence-transformers

Reason: The sentence-transformers library provides pre-trained BERT-based models that generate high-quality semantic embeddings for text. The all-MiniLM-L6-v2 model used in CareerBridge AI produces 384-dimensional vectors that enable accurate and efficient semantic similarity computation between resumes and job descriptions.

Library: spaCy

Reason: spaCy is an advanced industrial-strength NLP library used for Named Entity Recognition, tokenization, and custom vocabulary matching. In CareerBridge AI, spaCy powers the resume parsing module to extract candidate names, organizations, skills, and other key entities from uploaded PDF documents.

Library: PyMuPDF (fitz)

Reason: PyMuPDF is a high-performance PDF processing library that extracts raw textual content from uploaded resume

files. It handles various PDF formats reliably and serves as the first step in the resume analysis pipeline.
Library: TextBlob
Reason: TextBlob is a simple yet effective NLP library used for sentiment analysis tasks. In CareerBridge AI, it evaluates the tone and professionalism of resume content by computing polarity and subjectivity scores, providing candidates with feedback on their writing style.
Library: Pandas and NumPy
Reason: Pandas provides powerful data manipulation capabilities for handling structured datasets such as job listings, application records, and match score tables. NumPy supports the mathematical operations involved in vector computations, similarity calculations, and ATS score aggregation.
Toolkit: NLTK (Natural Language Toolkit)
Reason: NLTK supports supplementary NLP preprocessing tasks including stop word removal and tokenization during the text cleaning stage of the resume and job description processing pipeline.

4.2 Result:

The performance of Career Bridge AI was evaluated using standard information retrieval metrics — Precision, Recall, and F-Measure — on a test dataset comprising 50 candidate resumes evaluated against 20 distinct job descriptions, resulting in a total of 1000 resume-job pairs.
The F-Measure, also known as the F1-Score, is the harmonic mean of Precision and Recall and provides a balanced evaluation of model performance particularly when there is an uneven distribution of relevant and irrelevant matches.
The formula is defined as follows:
F-Measure = (2 × Precision × Recall) / (Precision + Recall)

Where Precision and Recall are defined as:
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)

Substituting into the F-Measure equation and simplifying:
F-Measure = 2TP / (2TP + FP + FN)

For the BERT-based semantic matching engine of Career Bridge AI, the values are calculated as follows:
Precision = Accurate Matches / (Accurate Matches + Wrong Matches)
Precision = 178 / (178 + 20)
Precision = 0.899
Recall = Accurate Matches / (Accurate Matches + Missed Matches)
Recall = 178 / (178 + 22)
Recall = 0.890
F-Measure = (2 × 0.899 × 0.890) / (0.899 + 0.890)
F-Measure = (2 × 0.800) / 1.789
F-Measure = 0.894

This result demonstrates that the BERT-based semantic matching engine of CareerBridge AI achieves both high precision and high recall, indicating that the system accurately identifies relevant job matches while minimizing both false positives and missed matches.

System	Precision	Recall	F-Score
Keyword Matching	0.610	0.640	0.624
TF-IDF + Cosine Similarity	0.740	0.760	0.749
CareerBridge AI (BERT-based)	0.899	0.890	0.894

Table 1. Performance Comparison of Matching Approaches

The results validate the efficacy of semantic embeddings for resume-job matching tasks by clearly showing that CareerBridge AI's BERT-based semantic matching performs significantly better across all three evaluation metrics than both keyword-based and TF-IDF-based approaches.

V.CONCLUSION

A comprehensive and clever solution to the enduring inefficiencies that define conventional recruitment systems is offered by CareerBridge AI. The platform significantly alters the way resumes and job descriptions are compared and matched by substituting BERT-powered semantic understanding for superficial keyword-based filtering. This allows for the accurate identification of relevant candidates even when different terminology is used to express equivalent qualifications.

All parties involved in the hiring process have their needs met by the system on a single, cohesive platform. The guesswork that usually surrounds application outcomes is eliminated for job seekers with clear and understandable match reports that show which skills match job requirements and which are lacking. Instead of applying mindlessly to positions for which they are unprepared, candidates are further empowered by the personalized learning roadmaps to take focused and informed steps toward enhancing their employability. The bidirectional matching engine eliminates the need for laborious manual screening by automatically ranking all registered candidates against each job posting. This results in an AI-curated shortlist for the recruiter. The platform greatly shortens the hiring process and raises the caliber of shortlisted applicants when combined with the HR analytics dashboard, skill coverage heatmaps, and bulk action features. By making the resume evaluation process transparent to candidates prior to application submission, the ATS scoring module solves a commonly disregarded issue in contemporary recruitment, enabling them to proactively optimize their documents. All competing platforms lack the sentiment analysis module's additional layer of qualitative feedback on the tone and professionalism of resume writing. The BERT-based matching engine significantly outperformed keyword matching and TF-IDF based methods, achieving a Precision of 0.899, Recall of 0.890, and F-Measure of 0.894 in an experimental evaluation on a dataset of 50 resumes across 20 job descriptions. These findings demonstrate that semantic comprehension is a crucial benefit in AI recruitment.

In conclusion, CareerBridge AI effectively illustrates how contemporary AI can be used to build a more equitable, intelligent, and open hiring environment. With upcoming improvements like real-time job scraping, conversational AI career assistants, deep learning fine-tuning on domain-specific recruitment datasets, and support for mobile applications, the system is well-positioned to develop into a fully autonomous career intelligence platform that more successfully connects talent and opportunity than any current solution.

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